



Virtual Member Meeting

July 2020

Logistics

- This meeting is being recorded.
- There is time today devoted to Q&A after the working group updates.
- Type your questions into the question pane throughout the meeting.
- If we don't have time for all questions, we will follow-up with you directly.
- The Option A/B meetings are the same meeting, agenda and slides. You do not need to attend both.

Agenda

Welcome and opening remarks

- Richard Soley, Executive Director
- Bill Ruh, Lendlease and Steering Committee Chair

Working Group Updates and Initial Projects

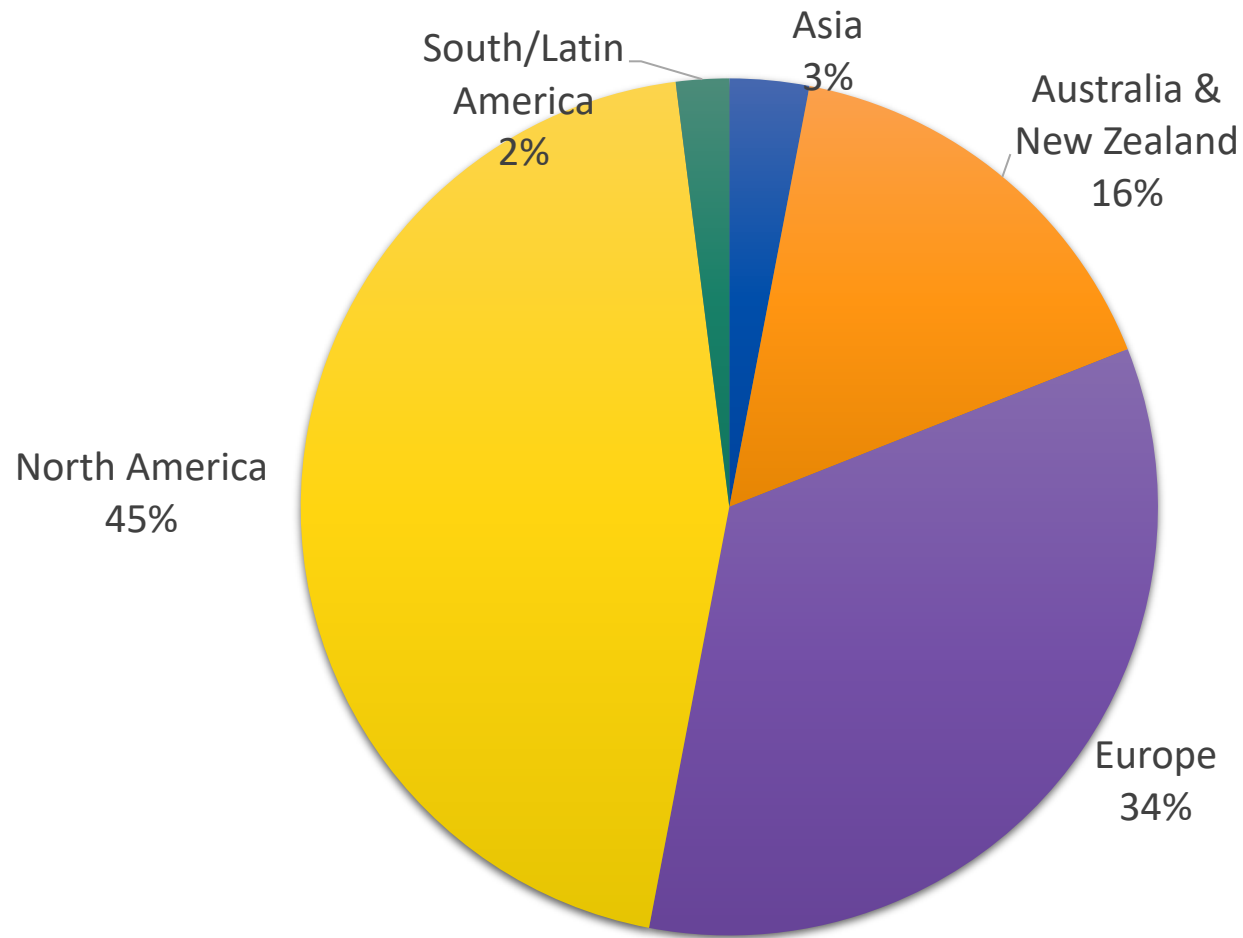
- Technology, Terminology and Taxonomy
- Infrastructure
- Natural Resources
- Manufacturing
- Aerospace & Defense
- Marketing

Q&A

Current members



Members by region



22 Countries Represented

Australia	Lithuania
Belgium	Netherlands
Brazil	New Zealand
Canada	Portugal
Chile	Russia
France	Spain
Germany	Sweden
Hong Kong	Switzerland
India	Turkey
Ireland	U.S.A.
Israel	United Kingdom

Steering Committee



Maturity of Digital Twin Market

13%

organizations that have
already used digital twins

62%

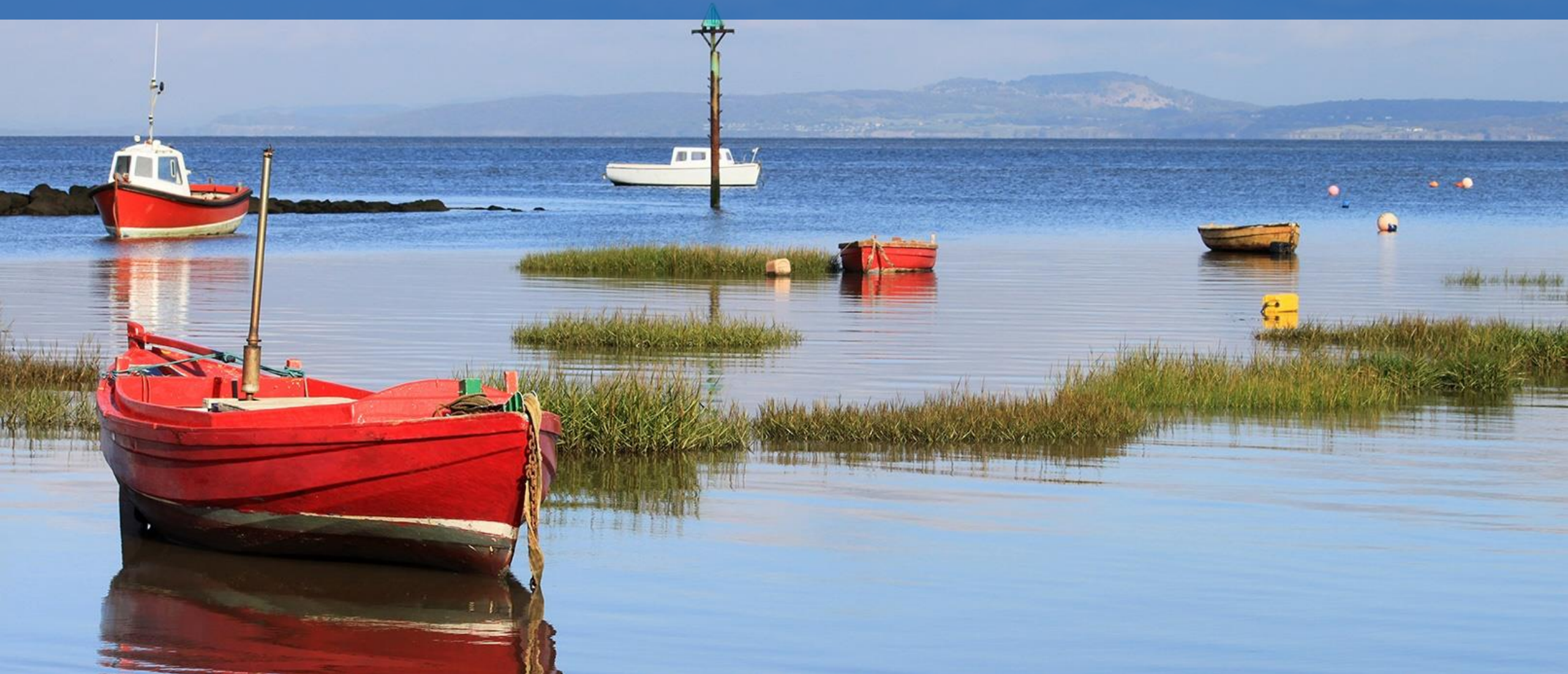
organizations in the
process or plan to use
digital twins within 1 year

67%

companies implementing
IoT will have at least one
digital twin in production
by 2022

Source: 2019 IoT implementation survey by Gartner

Drive the market, accelerate the value



Initial Working Groups



Technology,
Terminology &
Taxonomy



Infrastructure



Manufacturing



Natural Resources



Defense & Aerospace



Marketing

We can't do everything at once. We need to remain focused and prioritize projects.



First-year Priorities

Develop use
case library

Define open
source
implementation

Define
terminology
& taxonomy

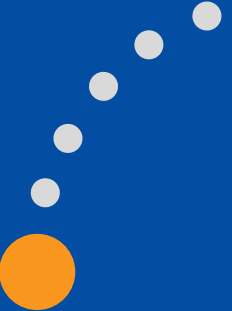
Identify Types
of Digital
Twins:
Process,
Simulation,
etc.

Identify Product
Lifecycle: Design,
As Built, As
Manufactured,
As Maintained

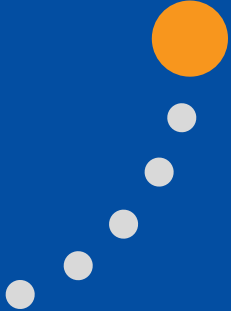
Common
Challenges
vs. Industry
Specific
Vertical

Establish Global
Digital Twin
Conference





Working Group Updates



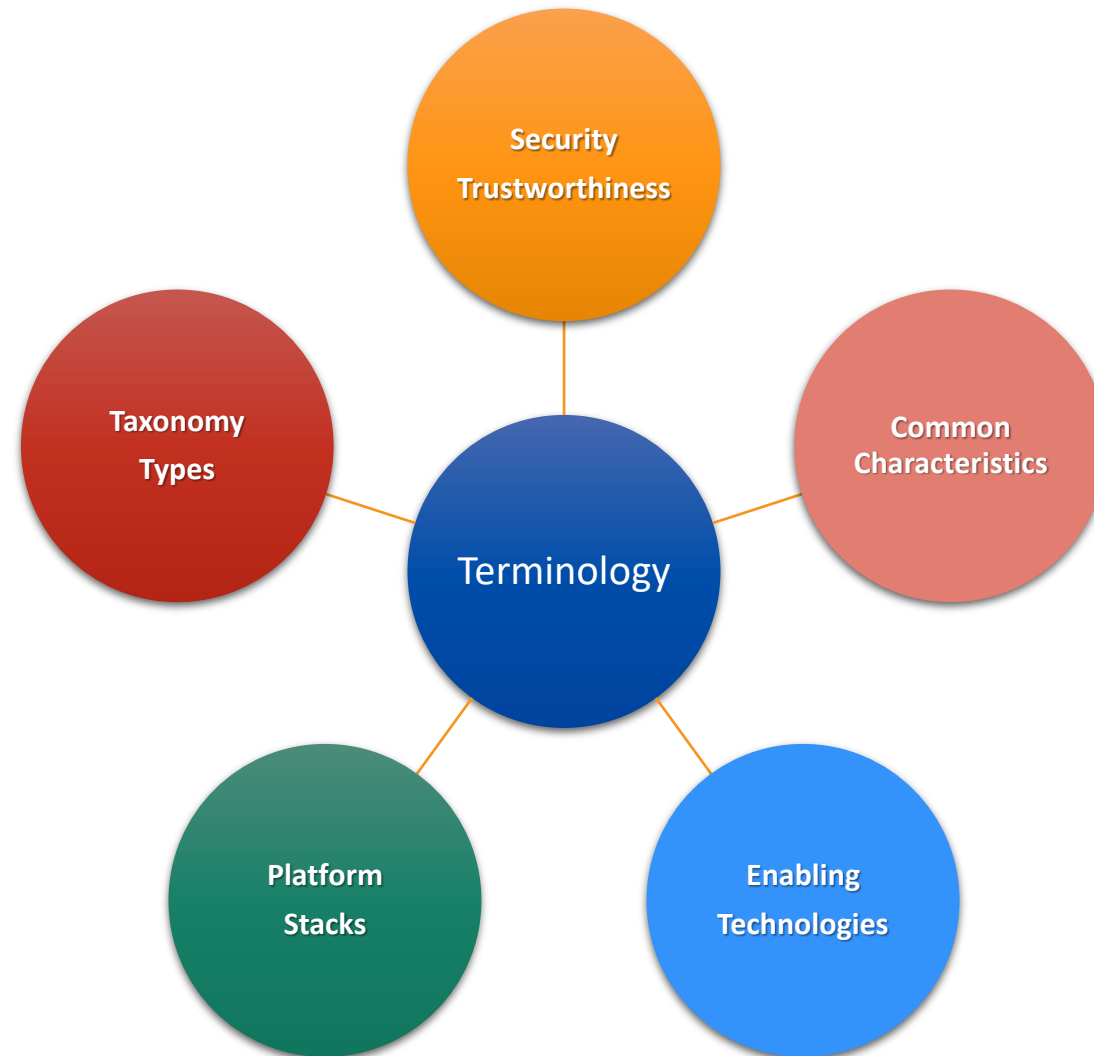
Technology, Terminology & Taxonomy WG



Technology,
Terminology &
Taxonomy

- The Digital Twin Technology, Terminology & Taxonomy Working Group will recommend a preferred definition, taxonomy, and, ultimately, hierarchy of “Digital Twins” to enable the industry to speak with a common vocabulary on this concept – enabling better understanding through shared definition and vernacular.
- Chair: Said Tabet (Dell) – said.tabet@dell.com
- Co-chair positions open!
- Next meeting: Thursday, July 9 at 10am ET

Technology, Terminology & Taxonomy



First-year Priorities

Develop use
case library

Define open
source
implementation

**Define Technology
Terminology &
Taxonomy
Security and
Trustworthiness**

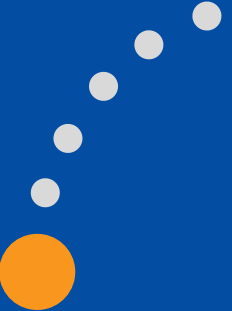
**Identify
Digital Twin
Types**
Process,
Simulation, etc.

**Identify
Product Lifecycle**
As Built, As
Manufactured, As
Maintained

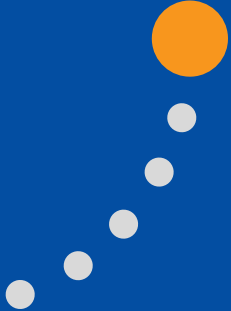
Common
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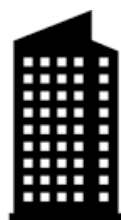


Infrastructure WG



Infrastructure Working Group

- Co-Chairs – Richard Ferris (Lendlease)
- Salla Eckhardt (Microsoft)
- John Turner – (Gafcon)
- Create more liveable, workable, and sustainable communities
 - Reduce cost – efficiency in how we design, build and operate infrastructure
 - Use data to deliver improved services
 - Sustainable promoting circular economy - the way we produce, assemble, sell and use products to minimise waste, and to reduce our environmental impact.
 - Socially inclusive



List of Current Members – Infrastructure (50)

- ECS Global Solutions
 - Lendlease
 - Jitsuin
 - NSW State Government
 - Microsoft
 - Trendspek
 - IOTIFY
 - e-Magic
 - Lux Modus
 - Bentley
 - ANSYS
 - Twin Building
 - Piprate
 - Bandora Systems
 - Dell
 - YoGeo
 - IoT Management
 - Gafcon
 - Idun Real Estate Solutions
 - Ricardo
 - Dell
 - LINQ
 - XMPro
 - Monash University
 - PropTechNL
 - ConstruWise
 - WSC Technology
 - DigitalTwin Technology GmbH
 - Cibersur Lda.
 - Willow
 - Geminus.AI
 - BIM6D Consulting
 - LUNO UAB
 - Autiosalo
 - Building 4.0 CRC
 - IMEC
 - Padi LLC
 - BEC - Blockchain Engineering Council
 - Berkeley & Imperial LP
 - Animated Insights
 - UrsaLeo
 - HyBird
 - dataArrows
 - Numondial Digital Telecom Svcs
 - Ricardo
 - OEMG
 - Arendai
- Questions: We are scheduling presentations
 - Use Cases / technology showcases /
 - Promotional Material – Events - Maureen Robusto (robusto@digitaltwinconsortium.org)

Workgroup Activities \ Volunteers

- Teams working adapting the Digital Twin Lifecycle to Identify/Design, Delivery, and Operations.
- Use Cases being prepared
- Great Cadence on Use Case Presentations
 - Microsoft - Digital Twin means to Microsoft through Design, Construction and Operations
 - OEMG - Digital Twin for the Sydney Harbour crossing for the Metro
 - Luxmodus - Digital twin for transmission pipelines and how they are used during construction?

Up - Coming

- Centre for Spatial Data Infrastructures and Land Administration (CSDILA) at the University of Melbourne
- iotify - simulating occupancy sensors in a large multi-story building and modelling the flow of people in a pseudo-realistic environment.
- NSW State Government - Spatial Digital Twin

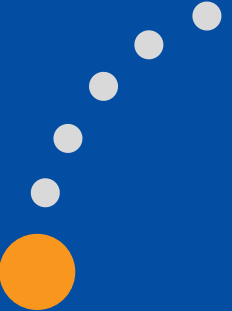
Join Us – Infrastructure WG

Use Case Presentations

- July 16 (US) – 5pm ET
- July 17 (Australia) – 7am AEST

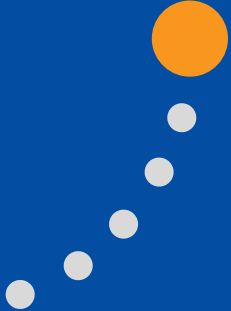
Working Group Meeting

- July 21 (US) – 5pm ET
- July 22 (Australia) – 7am AEST
- Meets every other Tuesday
(opposite weeks from Marketing WG)



Natural Resources WG

Co-chairs:

- Ron Zahavi (interim) – Microsoft
 - Pieter van Schalkwyk – XMPro
- 

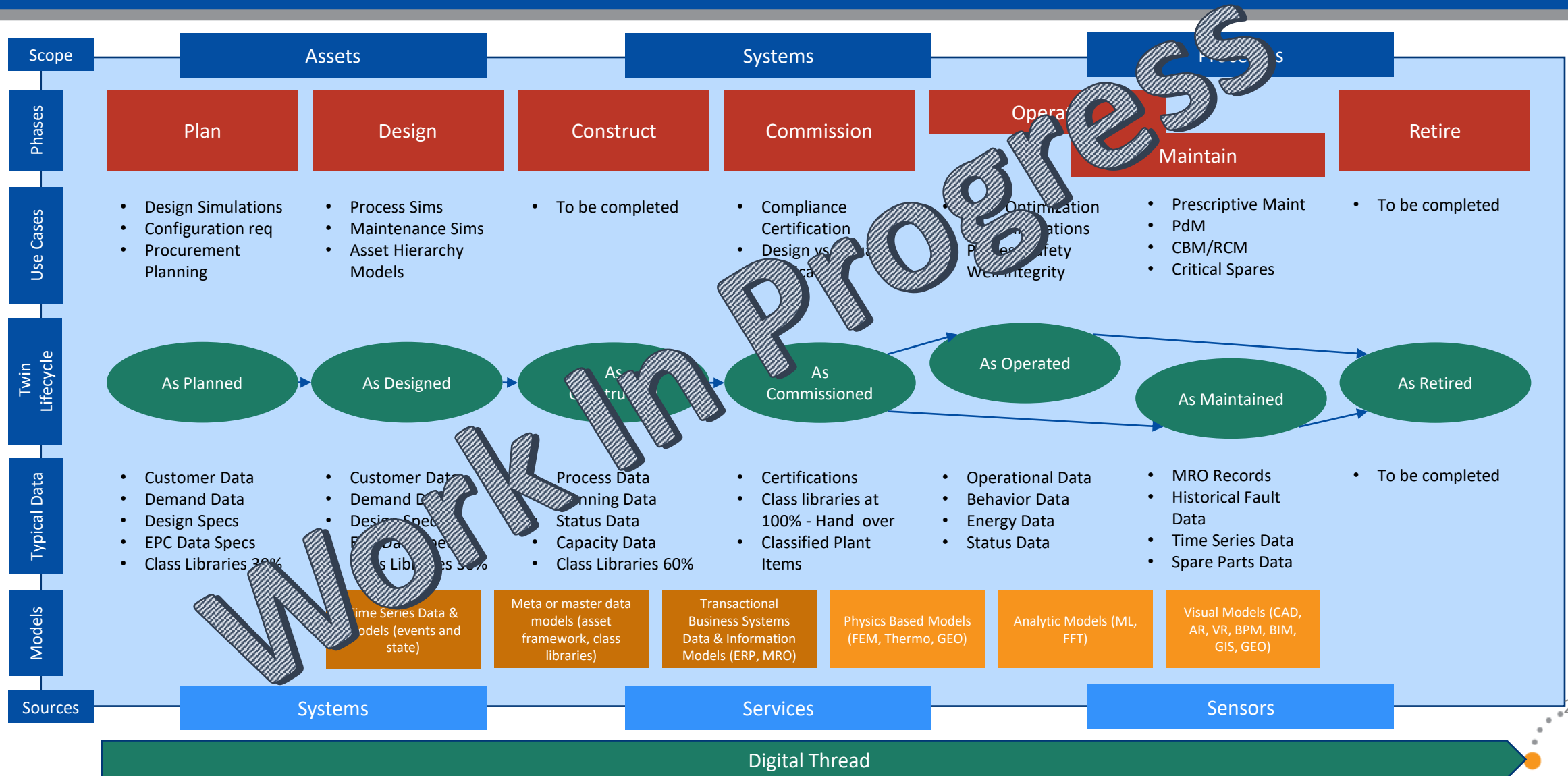
Charter

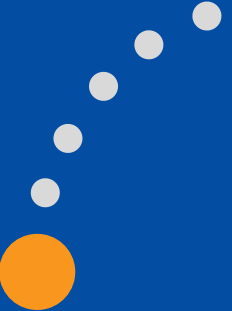
Natural resources related sectors, including Oil&Gas and mining, are becoming increasingly digital and digital twins can be applied during the engineering, design, construction and deployment, and operations phases of the lifecycle to improve production and predict or detect problems as well as improve safety. The natural resources digital twin working group will focus on the applicability of digital twin to the upstream Oil&Gas processes as well as optimize mining operations. Activities will include

- Applicability of digital twins to different stages of the production lifecycle
- Exploration of commonality and differences between different types of natural resources industries
- Identify the need for digital twin natural resources interoperability
- Look at the interaction required between physical and digital production components and digital feedback loops
- Define semantics required to describe a natural resources or subsection of industries digital twin
- Identify standards gaps and requirements
- Identify opportunities for testbed activities
- Collaborate with other DTC groups such as vocabulary to define a natural resources set of digital twin taxonomy, with liaison group to identify natural resources industry organizations for collaboration, and define reference architecture
- The working group will publish opinions and position statements related to regulations and policies, define and launch testbeds, identify standards gaps, create a semantic model and taxonomy, and a reference architecture

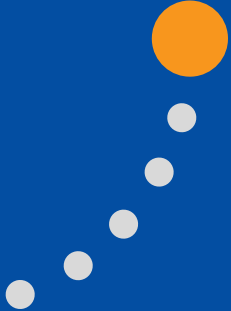
- EXAMPLES OF WORK -
Proposed Lifecycle Model
To Classify Digital Twin Use Cases

Digital Twin Life Cycle in Natural Resources





Develop Use Case



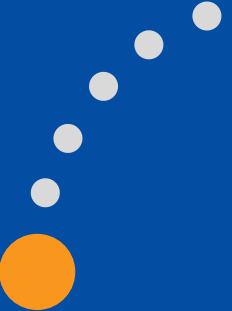
Use Case List (example)

Oil & Gas

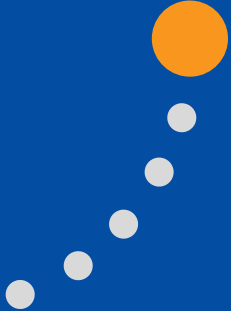
- Offshore Well Integrity Digital Twin (operate)
- EPC Digital Twin manage classified EEHA equipment plus Vessels and PSV's (Design and Construction)
- Simulation Twin to lower risk with virtual commissioning, training, and startup prior to launch (commissioning – refinery)
- Digital Twin to Optimize over- and underwater systems (Operate)
- Simulation Twin to reduce the time needed to estimate project parameters by using digital simulators (design)
- Simulation twin to increase the accuracy of cost-weight estimations by using integrated error-proof solutions (design)
- Gas Turbine Twin for Predictive Maintenance (maintain)

Mining

- Digital Twin for Coal Handling and Processing Plant (CHPP) during Operations
- Autoclave Digital Twin for Gold Processing (Operate)
- Long Conveyor Predictive Maintenance Digital Twin (Maintain)

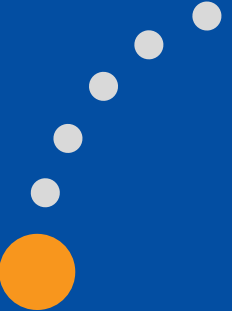


Use Case Descriptions

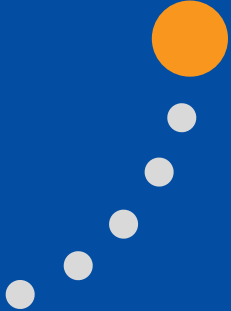


Mining (example)

Use Case Name Digital Twin for Coal Handling and Processing Plant (CHPP)	Industry(ies) Mining
Use Case Description: CHPP Throughput Loss Management Coal washeries can account for 20 – 30% mass loss through the process of mineral matter from coal and this impacts washery yield. Supervisors and General Managers only find out about losses through weekly reports. At a coal price of \$100/tonne, a 1% yield loss due to inefficient operation of this circuit would represent \$6 million per year in lost sales.	Lifecycle Operate
	Objective Classification Production Optimization
Scenario Description A digital twin for a CHPP plant monitor real-time operation data feeds from sensor and control systems. The data is used as an input to a digital metallurgical/process model that provides real-time event data on losses per shift. The digital twin further provide simulation capabilities using the real-time data and simulated process settings to find ideal process parameters to reduce washing losses. The digital twin notify shift losses to supervisors and mine managers for appropriate response	Roles/Actors Personas Production Operators, Supervisors, and General Manager/Mine Manager
	Special Requirements

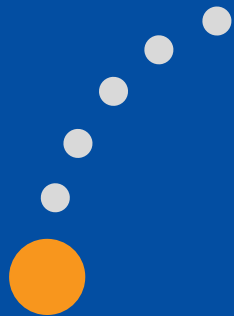


Identify Industry Standards and Specifications in use - Information Models -



Specifications and standards in use

- [ISO 64076](#) Petroleum, petrochemical and natural gas industries -- Collection and exchange of reliability and maintenance data for equipment
- [ISO 15926](#) is a standard for data integration, sharing, exchange, and hand-over between computer systems.
- [ISO 10303](#) is an ISO standard for the computer-interpretable representation and exchange of product manufacturing information. Its official title is: Automation systems and integration — Product data representation and exchange. It is known informally as "STEP", which stands for "Standard for the Exchange of Product model data". ISO 10303 can represent 3D objects in Computer-aided design (CAD) and related information.
- [PRODML](#) is a family of XML and Web Services based upstream oil and natural gas industry standards from Energistics and its PRODML Special Interest Group (PRODML SIG).
- [WITSML](#) - Covering Drilling, Completions and Interventions, WITSML is the industry reference for the transmission of data from rig-site to the offices of oilfield service companies, integrators and operators.
- [ISO 14224:2016](#) provides a comprehensive basis for the collection of reliability and maintenance (RM) data in a standard format for equipment in all facilities and operations within the petroleum, natural gas and petrochemical industries during the operational life cycle of equipment.
- [RESQML](#) is an industry initiative to provide open, non-proprietary data exchange standards for the reservoir life cycle, from initial structural interpretation and modeling, to reservoir characterization, static and dynamic simulation, and production monitoring.
- [OSDU](#) – Open Subsurface Data Universe for standardized data services, storage, and APIs in Oil & gas



Next Steps



Current Activities

- Collect Use Case list across different Natural Resources industries
- Classify by Digital Twin lifecycle stage
- Expand and publish Use Cases
- Extract common elements for reference architecture
- Identify scope for open source
- Collaborate with Taxonomy & Terminology WG on lifecycle descriptions and classifications
- Collaborate with other WGs to standardize approach

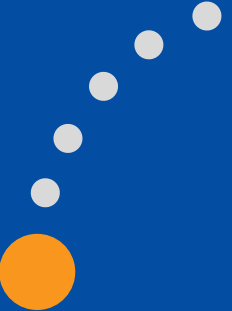
Get Involved

- Join the Working Group in Causeway – Natural Resources
- Join our Working Group calls (every second Thursday US)
- Contribute to the use cases and other work items

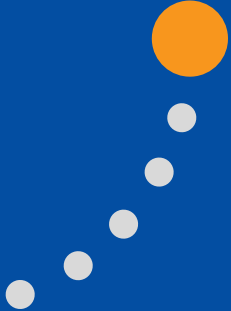
Co-chairs: Get in touch if you have any questions

- Pieter van Schalkwyk – pvs@xmpro.com
- Ron Zahavi – ron.zahavi@microsoft.com





Manufacturing WG

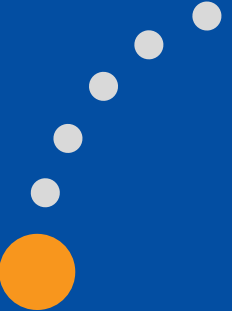


Manufacturing WG

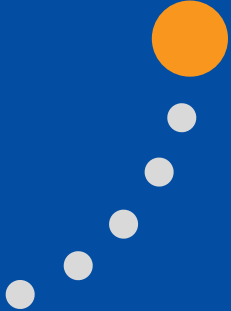


Manufacturing

- Manufacturing is becoming increasingly digital and digital twins can be applied during the engineering, design, production and operations phases of a product lifecycle to improve products and predict or detect problems. The manufacturing digital twin working group will focus on the applicability of digital twin to the manufacturing process in various industries.
- Co-chairs:
 - Sameer Kher (Ansys) - sameer.kher@ansys.com
 - Todd Edmunds (Dell) - todd_edmunds@dell.com
 - Lee Johnson (Geminus) - lee@geminus.ai
- Next meeting: Tuesday, July 14 – 10am ET
 - Tuesday, July 28 – 4pm ET
 - Every other Tuesday; alternating at 10 am ET and 4 pm ET



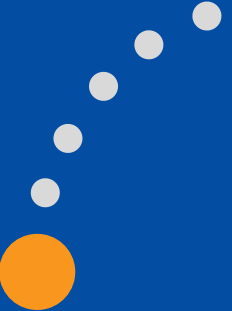
Aerospace & Defense WG



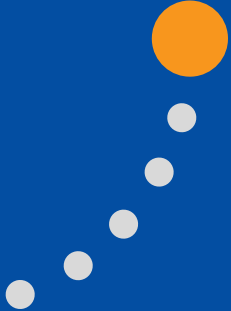
Defense & Aerospace WG



- Defense and Aerospace companies and Government Agencies have been early adopters of Digital Twins with significant usage already in the operation phase. As an example, the USAF has significant Condition Based Monitoring (CBM+) initiatives.
- The Defense and Aerospace Digital Twin working group will focus on the applicability of Digital Twins across the lifecycle in various industries.
- Co-chairs
 - Jordan Garrett (Dell) - jordan.garrett@dell.com
 - James Sumpter (US Air Force) - james.sumpter.1@us.af.mil
- Meetings
 - Every other Wednesday at 12:00pm ET



Marketing WG



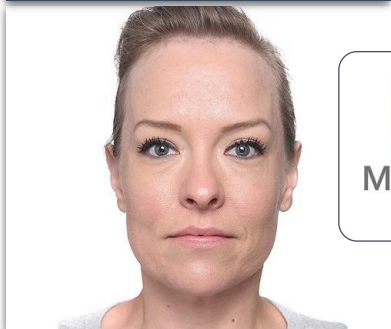
Marketing Working Group

The mission of the Marketing Working Group is to position Digital Twin Consortium as the Authority in Digital Twin.

- **Objectives**

- Improve brand awareness and perception of Digital Twin Consortium
- Encourage membership growth and retention
- Increase adoption of Digital Twin Consortium guidelines/publications

Salla Eckhardt



Maureen Robusto



Josh Adams



Chris Hussey



Marketing opportunities

	Press releases
	Media & analysts
	Webinars
	Social media
	Videos
	Events
	Speakers bureau

- Participate
 - Authors get interviews & webinars
- Tag us
 - We'll share posts as long as there is no appearance of us supporting a specific product/service
- Member Newsroom
 - Send us [Karen] your press releases and we'll post
- Member Events
 - Send us [Carol] your events for our member events calendar

Marketing resources in Causeway

The screenshot shows a web browser window with the URL `engage.digitaltwinconsortium.org/wg/Marketing/dashboard`. The browser's address bar and tabs are visible at the top. Below the browser window, the dashboard interface is displayed. It features a top navigation bar with links like 'Welcome, Maureen', 'My Profile', 'Public Site', and 'Sign Out'. A search bar is also present. The main navigation bar includes 'Marketing' and a dropdown menu. Below this, there are tabs for 'Dashboard', 'Documents', 'Discussions', and 'Calendar'. The dashboard content is organized into several panels:

- Marketing**: A sidebar panel listing resources such as 'Word template and letterhead', 'Webinar guidelines', 'PPT template', 'Sales deck', 'Brand guide', 'Membership Badge (png)', 'Membership Badge (eps)', and 'Logo Request Form'.
- Marketing Discussions**: A panel showing a list of discussions. The first two are 'Event "Member Meeting (option B)" has been...' and 'Event "Member Meeting (option A)" has been...', both dated Jul 1, 2020, at 3:13 PM, by Maureen Robusto. The third is '2020-06-23 Marketing Working Group Meeting Minutes' dated Jun 25, 2020, at 3:29 PM, by Michael Boldt. There are buttons for 'Post a Message' and 'View Archives'.
- Marketing Calendar**: A panel showing a list of events. The first two are 'Member Meeting (option A)' and 'Member Meeting (option B)', both on Jul 8, 2020, at 9:00 AM and 6:30 PM respectively. The third is 'Marketing WG Digital Twin 2nd Tues' on Jul 14, 2020, at 9:00 AM. There are buttons for 'Add Event' and 'View Calendar'.
- Marketing Documents**: A panel showing a list of documents. The first is 'Badge (.png) - Member since 2020.png' dated Jun 30, 2020. The next two are '2020-06-23 Digital Twin Marketing Working Group...' dated Jun 25, 2020, Rev 0. There is a 'Download' button for each document and an 'Upload Document' button at the bottom.
- Workgroup Details**: A panel showing the 'Workgroup Chair' as Joshua Adams with email `josh.adams@animated-insights.com`. It also lists other members: Salla Eckhardt, Chris Hussey, Maureen Robusto, and Tracie Berardi.

Marketing Staff



Maureen Robusto

- Messaging
- Website
- Content
- Sales enablement
- robusto@



Karen Quatromoni

- Press releases
- Media & analysts
- Speakers bureau
- karen@



Meg Duncan

- Social media
- Webinars
- Member newsletters
- Lead generation
- meg@



Carol David

- Tradeshows
- Member meetings
- carol@



Evan Birkhead

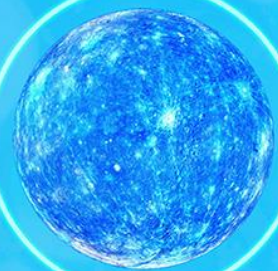
- Marketing affiliates
- Content
- birkhead@

Our emails: @digitaltwinconsortium.org

Join us – Marketing WG

- Tuesday, 14 July
 - 6am Seattle
 - 9am Boston
 - 2pm London
 - 11pm Sydney
- Tuesday, 28 July
 - 2pm Seattle
 - 5pm Boston
 - 10pm London
 - 7am Sydney (Wednesday, 29 July)
- Meetings every other Tuesday
- Opposite weeks as Infrastructure WG
- Alternating times – 9am ET one week; 5pm ET two weeks later

Q & A



Final comments

- Onboarding next week with Diane & Dan
 - Thursday, July 16 at 11:00 am EDT – [Register](#)
 - Thursday, July 16 at 7:00 pm EDT – [Register](#)
- Or email the helpdesk anytime
 - help@digitaltwinconsortium.org



Diane Ehamjian
Director Member
Engagement



Dan Isaacs
VP, Technical Director